

Gestation Guardian

A Mobile-First Maternal Health Monitoring Application

Abstract

Gestation Guardian is a mobile-first maternal health monitoring application designed to assist expectant mothers in tracking vital health parameters throughout pregnancy. Built as a hybrid web application using TypeScript, Vite, and an Android WebView wrapper, the app provides real-time health tracking, risk assessment, and educational guidance within a single-page application architecture.

The application features a comprehensive dashboard displaying gestational age, trimester progression, and a visual progress tracker calculated from the user's last menstrual period (LMP). Users can log blood pressure readings with systolic and diastolic values, and record daily vitals including weight, sleep hours, stress levels, body temperature, blood glucose, and urine protein levels. A dedicated fetal kick counter with a timed session interface allows mothers to monitor fetal movement patterns, while a separate contraction timer tracks the duration and intervals between contractions.

At the core of the application is a risk scoring engine that evaluates the user's health state by combining static risk factors (maternal age, parity, prior preeclampsia history, chronic hypertension, diabetes status, family history, multiple gestation, and BMI) with dynamic signals (blood pressure readings, proteinuria levels, blood glucose values, symptoms such as severe headache, visual disturbances, right upper quadrant pain, sudden swelling, and trend-based weight gain analysis). The engine produces a numerical risk score and classifies the result into a Red-Amber-Green (RAG) triage band: Low, Moderate, High, or Critical. Each band triggers a clinical action recommendation displayed on the dashboard.

The data layer uses a hybrid storage architecture with Firebase Cloud Firestore as the primary backend and browser localStorage as a synchronous fallback. The application detects whether valid Firebase credentials are configured and gracefully degrades to offline-only operation when they are not. IndexedDB persistence is enabled for offline caching when cloud connectivity is available.

An integrated AI chatbot provides contextual pregnancy guidance through a keyword-matching mock engine that responds to queries about nutrition, iron intake, sleep positions, fetal kick counts, blood pressure monitoring, contractions, labor signs, and nausea management. The chatbot features a draggable floating action button, a full-screen chat overlay with a typing indicator, and asynchronous response delivery.

Additional modules include a medical history recorder, a health records hub aggregating logged data across all vitals categories, a care guide with trimester-specific educational content, a reminder system, user authentication with profile management, and a Bluetooth device integration stub for future connectivity with blood pressure cuffs and fetal dopplers via the Web Bluetooth API. The application is compiled using Vite with a legacy polyfill plugin and packaged as an Android APK through a Kotlin-based WebView activity.

Keywords

Maternal health, pregnancy monitoring, preeclampsia risk scoring, RAG triage, fetal kick counter, contraction timer, blood pressure tracking, hybrid mobile application, Firebase, TypeScript, Android WebView

Technology Stack

- Frontend: TypeScript, HTML5, CSS3 (Single Page Application)
- Build Tool: Vite 8.1.4 with Legacy Polyfill Plugin
- Backend: Firebase Cloud Firestore with IndexedDB offline persistence
- Local Storage: Browser localStorage (synchronous fallback)
- Mobile: Android WebView (Kotlin, Gradle)
- Testing: Playwright
- AI Module: Keyword-matching mock engine (mock-ai.ts)

Application Modules

- Dashboard: Gestational age, trimester display, progress bar, baby size indicator, RAG risk banner
- Blood Pressure Logger: Systolic/diastolic input with position tracking (sitting, lying, standing)
- Vitals Logger: Weight, sleep, stress, temperature, blood glucose, urine protein
- Fetal Kick Counter: Timed session with movement count and target tracking
- Contraction Timer: Duration and interval tracking with session history
- Risk Scoring Engine: Static and dynamic factor analysis with RAG triage classification
- AI Chatbot: Contextual pregnancy guidance with typing indicator and draggable FAB
- Health Records Hub: Aggregated view of all logged health data by category
- Care Guide: Trimester-specific educational content and clinical guidance
- Medical History: Obstetric and general medical history recorder
- Profile and Authentication: User registration, sign-in, and profile management
- Reminders: Configurable appointment and medication reminders
- Bluetooth Integration: Stub for blood pressure cuff and fetal doppler connectivity